SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	N.Sai Sankar	**Reg. No.:**	192311187
Section / Batch:	CSE	**Date:**	6/10/2026

Code of Conduct

I N.Sai Sankar(192311187) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.Sai Sankar

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

ANSWERS

Question:1

Problem Understanding

A vacuum-cleaner agent operates in a two-square environment containing squares A and B. The agent can perceive its current location and whether that square is Dirty or Clean.

- Initial position: A
- Initial condition: A = Dirty, B = Dirty
- Actions: Suck, Left, Right, NoOp
- Lifetime: T = 6 time steps
- Reward: +1 for every clean square at the end of each step
- Movement penalty: p = 1 for every Left or Right movement
- Dirt does not reappear

Two agents are considered:

- Agent X: Simple reflex agent
- Agent Y: Reflex agent with internal state

(a) PEAS Description

PEAS Component	Description
Performance Measure	+1 for every clean square at the end of each time step; penalty of p for every Left/Right movement
Environment	Two squares A and B containing dirt; dirt does not reappear
Actuators	Suck, Left, Right, NoOp
Sensors	Sensor detects the agent's current location and whether the current square is Dirty or Clean

Final PEAS

P – Performance: Keep squares clean while minimizing movement penalty.

E – Environment: Two-room vacuum world with squares A and B.

A – Actuators: Suck, Left, Right and NoOp.

S – Sensors: Detect current location and dirt status.

(b) Trace Agent X for 6 Time Steps

Agent X Rule

- If current square is Dirty → Suck
- Otherwise → Move to the other square

Initial state:

A = Dirty, B = Dirty

For each step:

Step Score = Number of clean squares − Movement penalty

Trace Table

Time Step	Location	Percept	Action	Clean Squares at End	Move Penalty	Step Score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	Left	2	1	1
5	A	Clean	Right	2	1	1
6	B	Clean	Left	2	1	1

Total Score of Agent X

$$Score_X = 1 + 0 + 2 + 1 + 1 + 1$$

$$Score_X = 6$$

Therefore, Agent X obtains a total score of 6 when p = 1.

(c) Trace Agent Y for 6 Time Steps

Agent Y Rule

- If current square is Dirty → Suck
- If both squares are remembered as clean → NoOp
- Otherwise → Move to the other square

Trace Table

Time Step	Location	Percept	Action	Clean Squares at End	Move Penalty	Step Score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	NoOp	2	0	2
5	B	Clean	NoOp	2	0	2
6	B	Clean	NoOp	2	0	2

Total Score of Agent Y

$$Score_Y = 1 + 0 + 2 + 2 + 2 + 2$$

$$Score_Y = 9$$

Therefore, Agent Y obtains a total score of 9 when p = 1.

(d) Derive Total Score as a Function of p

Agent X

The total clean-square rewards for Agent X are:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Agent X makes 4 movements.

Therefore, total movement penalty = 4p

Hence,

$$Score_X = 10 - 4p$$

For p = 1:

$$Score_X = 10 - 4 = 6$$

Therefore,

$$Score_X = 10 - 4p$$

Agent Y

The total clean-square rewards for Agent Y are:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Agent Y makes only 1 movement.
Therefore, total movement penalty = p
Hence,

$$\text{Score_Y} = 10 - p$$

For p = 1:

$$\text{Score_Y} = 10 - 1 = 9$$

Therefore,

$$\text{Score_Y} = 10 - p$$

When is Y strictly better than X?
We require:

$$\text{Score_Y} > \text{Score_X}$$

Substituting the formulas:

$$10 - p > 10 - 4p$$

Subtract 10 from both sides:

$$-p > -4p$$

Add 4p to both sides:

$$3p > 0$$

Therefore,

$$p > 0$$

So, Agent Y is strictly better than Agent X whenever the movement penalty p is greater than 0.
At p = 0, both agents obtain the same score.

(e) Which Agent is Rational?
Agent Y is rational under the given performance measure.
The reason is that rationality depends on selecting an action that maximizes the expected performance based on the available percept sequence and internal information.
Agent X only reacts to the current percept. After both squares become clean, it does not remember that the other square has already been cleaned. Therefore, it continues moving between A and B, causing unnecessary movement penalties.
Agent Y has an internal state. It remembers that both squares have already been cleaned and chooses NoOp instead of repeatedly moving.
Therefore:

$$\text{Score_Y} = 10 - p$$

while

$$\text{Score_X} = 10 - 4p$$

For any p > 0,

$$\text{Score_Y} > \text{Score_X}$$

Interpretation
The example shows that percept history and internal state can improve rational decision-making. Both agents receive the same immediate percepts, but Agent Y uses information from previous percepts. Hence, it can avoid unnecessary actions and achieve a higher performance score.
This demonstrates why a rational agent may need more than a simple reflex rule.

Question:2

Task Environment Classification

The three environments are:

- E1: Autonomous taxi in city traffic
- E2: Part-picking robot using a camera on a moving conveyor belt
- E3: Chess-playing agent with a clock

The six properties are evaluated using the difficulty convention given in the question.

(a) Classification of Each Environment

E1 – Autonomous Taxi

Property	Classification	Justification
Observability	Partially observable	Sensors cannot observe everything
Agents	Multi-agent	Other vehicles, pedestrians and drivers interact
Determinism	Stochastic	Traffic and events are uncertain
Episodic/Sequential	Sequential	Current actions affect future driving
Static/Dynamic	Dynamic	Traffic conditions continuously change
Discrete/Continuous	Continuous	Position, speed and time vary continuously

E1 Result

Partially observable, multi-agent, stochastic, sequential, dynamic and continuous.

E2 – Conveyor-Belt Part-Picking Robot

Property	Classification	Justification
Observability	Partially observable	Camera may have limited view/occlusion
Agents	Single-agent	The robot performs the assigned picking task
Determinism	Stochastic	Part position and sensing/picking can be uncertain
Episodic/Sequential	Sequential	Picking actions occur continuously over the task
Static/Dynamic	Dynamic	Conveyor and parts are moving
Discrete/Continuous	Continuous	Position and movement are continuous

E2 Result

Partially observable, single-agent, stochastic, episodic, dynamic and continuous.

E3 – Chess-Playing Agent

Property	Classification	Justification
Observability	Fully observable	Complete chess board is visible
Agents	Multi-agent	Two players compete
Determinism	Deterministic	Legal moves have predictable board results
Episodic/Sequential	Sequential	Each move affects future moves
Static/Dynamic	Semi-dynamic	Board remains fixed while thinking, but clock continues
Discrete/Continuous	Discrete	Moves and board states are discrete

E3 Result

Fully observable, multi-agent, deterministic, sequential, semi-dynamic and discrete.

(b) Difficulty Score

According to the question:

- Harder property = 1
- Easier property = 0
- Semi-dynamic = 0.5

E1 – Autonomous Taxi

1+1+1+1+1+1=6

E2 – Part-Picking Robot

1+0+1+1+1+1=5

E3 – Chess

0+1+0+1+0.5+0=2.5

Ranking

Rank	Environment	Difficulty
1	E1 – Autonomous Taxi	**6**
2	E2 – Part-Picking Robot	**5**
3	E3 – Chess	**2.5**

Therefore:

{E1>E2>E3}

Hence, the ranking from hardest to easiest is:
E1 → E2 → E3

(c) Agent Architecture Selection

Hardest Environment – E1

E1 is the autonomous taxi environment.
A utility-based agent is suitable because the taxi must make decisions under:

- Partial observability
- Uncertainty
- Multiple agents
- Continuous changes
- Sequential consequences

The agent can evaluate different actions according to utility, such as safety, travel time and route quality.
Suitable architecture: Utility-based agent

Easiest Environment – E3

E3 is the chess environment according to the given difficulty score.
A goal-based agent is suitable because the agent has a clear goal:
Win the chess game.

It can evaluate possible moves and select actions that help achieve the goal.
Suitable architecture: Goal-based agent

(d) Interpretation

The complexity of an AI agent depends not only on the application domain but also on the properties of its task environment.

For example, an autonomous taxi operates in a partially observable, stochastic, dynamic and continuous environment. Therefore, it requires more sophisticated decision-making.

Chess is fully observable and deterministic, although it is sequential and multi-agent. Therefore, its environment has fewer difficult properties according to the given scoring system.

Question:3

The experiment contains 50 interrogations for each system. The number of times the judge correctly identifies the machine is given in the question.

System	Correctly Identified	Correct Identification Rate	Fooling Rate
ELIZA	47/50	94%	6%
Chatbot B	36/50	72%	28%
System C	29/50	58%	42%

(a) Correct-Identification Rate and Fooling Rate

Formula

Correct-identification rate:

$$\text{Correct Identification Rate} = (\text{Correctly identified} / \text{Total interrogations}) \times 100$$

Fooling rate:

$$\text{Fooling Rate} = 100 - \text{Correct Identification Rate}$$

ELIZA

Correct-identification rate:

$$(47/50) \times 100 = 94\%$$

Fooling rate:

$$100 - 94 = 6\%$$

Therefore:

ELIZA: Correct = 94%, Fooling = 6%

Chatbot B

Correct-identification rate:

$$(36/50) \times 100 = 72\%$$

Fooling rate:

$$100 - 72 = 28\%$$

Therefore:

Chatbot B: Correct = 72%, Fooling = 28%

System C

Correct-identification rate:

$$(29/50) \times 100 = 58\%$$

Fooling rate:

$$100 - 58 = 42\%$$

Therefore:

System C: Correct = 58%, Fooling = 42%

(b) Turing's 70% Criterion

The reference says that after five minutes, an average interrogator would have no more than a 70% chance of

correctly identifying the machine. Therefore, a system that produces a correct-identification rate of 70% or lower meets this criterion.

A pure guess gives a 50% correct-identification rate.

ELIZA

$$94\% > 70\%$$

Therefore, ELIZA does not satisfy the 70% criterion.

Distance from 50%:

$$94 - 50 = 44$$

Therefore:

44 percentage points above chance

Chatbot B

$$72\% > 70\%$$

Therefore, Chatbot B does not satisfy the 70% criterion.

Distance from 50%:

$$72 - 50 = 22$$

Therefore:

22 percentage points above chance

System C

$$58\% < 70\%$$

Therefore, System C satisfies the 70% criterion.

Distance from 50%:

$$58 - 50 = 8$$

Therefore:

8 percentage points above chance

Summary Table

System	Correct Identification	70% Criterion	Distance from 50% Chance
ELIZA	94%	Does not satisfy	+44 percentage points
Chatbot B	72%	Does not satisfy	+22 percentage points
System C	58%	Satisfies	+8 percentage points

(c) Improvement in Fooling Rate

ELIZA → System C

ELIZA fooling rate:

6%

System C fooling rate:

42%

Improvement:

$$42 - 6 = 36$$

Therefore:

36 percentage points

Ratio

$$42 / 6 = 7$$

Therefore, the fooling rate of System C is:

7 times the ELIZA fooling rate

Chatbot B → System C

Chatbot B fooling rate:

28%

System C fooling rate:

42%

Improvement:

$$42 - 28 = 14$$

Therefore:

14 percentage points

Final Answer

Comparison	**Improvement**
ELIZA → System C	36 percentage points
ELIZA → System C ratio	7 : 1
Chatbot B → System C	14 percentage points

(d) Interpretation

The Turing Test is concerned with whether a machine can act humanly during a conversation. A high fooling rate indicates that judges were more likely to mistake the machine for a human.

However, a high fooling rate does not prove that the system is thinking or acting rationally.

One important argument from the foundations of AI is the Chinese Room argument. It argues that a system may produce appropriate language responses by following rules without necessarily demonstrating genuine understanding of the meaning of those responses.

Therefore, System C's 42% fooling rate shows its performance in the particular conversational test, but it does not by itself establish that System C possesses human-like understanding, consciousness, or rationality.

Question:4

A taxi must reach the airport within a 55-minute deadline.
The utility function is:
$U = 100 - T - 20L$
where:

- T = travel time in minutes
- L = 1 if the taxi arrives after 55 minutes
- L = 0 if the taxi arrives within 55 minutes

The three available routes are:

Route	Normal Travel Time	Disruption Probability	Disrupted Travel Time
R1 – Highway	30 min	0.30	70 min
R2 – City	45 min	0.10	60 min
R3 – Ring Road	50 min	0	50 min

(a) Utility of Each Outcome
Utility Formula
$U = 100 - T - 20L$
If the taxi arrives within 55 minutes:
$L = 0$
If the taxi arrives after 55 minutes:
$L = 1$

R1 – Highway
Normal condition
Travel time:
$T = 30$
Since $30 \leq 55$:
$L = 0$
Therefore:
$U = 100 - 30 - 20(0)$
$U = 70$
So,
R1 Normal Utility = 70
Disrupted condition
Travel time:
$T = 70$
Since $70 > 55$:
$L = 1$
Therefore:
$U = 100 - 70 - 20(1)$
$U = 10$
So,
R1 Disrupted Utility = 10

R2 – City
Normal condition

Travel time:
$T = 45$
Since $45 \leq 55$:
$L = 0$
Therefore:
$U = 100 - 45 - 20(0)$
$U = 55$
So,
R2 Normal Utility = 55
Disrupted condition
Travel time:
$T = 60$
Since $60 > 55$:
$L = 1$
Therefore:
$U = 100 - 60 - 20(1)$
$U = 20$
So,
R2 Disrupted Utility = 20

R3 – Ring Road
Travel time:
$T = 50$
Since $50 \leq 55$:
$L = 0$
Therefore:
$U = 100 - 50 - 20(0)$
$U = 50$
So,
R3 Utility = 50

Utility Summary Table

Route	Outcome	Time	Late?	Utility
R1	Normal	30 min	No	**70**
R1	Disrupted	70 min	**Yes**	**10**
R2	Normal	45 min	No	**55**
R2	Disrupted	60 min	**Yes**	**20**
R3	Normal	50 min	No	**50**
R3	Disrupted	50 min	No	**50**

(b) Expected Utility of Each Route
Expected Utility Formula
$EU = P(\text{Normal}) \times U(\text{Normal}) + P(\text{Disruption}) \times U(\text{Disruption})$

R1 – Highway
Disruption probability = 0.30

Therefore:
P(Normal) = 1 − 0.30 = 0.70
Expected utility:
EU(R1) = (0.70 × 70) + (0.30 × 10)
EU(R1) = 49 + 3
EU(R1) = 52
Therefore:
EU(R1) = 52

R2 – City
Disruption probability = 0.10
Therefore:
P(Normal) = 1 − 0.10 = 0.90
Expected utility:
EU(R2) = (0.90 × 55) + (0.10 × 20)
EU(R2) = 49.5 + 2
EU(R2) = 51.5
Therefore:
EU(R2) = 51.5

R3 – Ring Road
Disruption probability = 0
Therefore:
P(Normal) = 1
Expected utility:
EU(R3) = (1 × 50) + (0 × 50)
EU(R3) = 50
Therefore:
EU(R3) = 50

Expected Utility Summary

Route	Expected Utility
R1 – Highway	52
R2 – City	51.5
R3 – Ring Road	50

Therefore, based on the given probabilities and utility function:
R1 – Highway has the highest expected utility of 52.
Rational Choice
The rational choice is R1 – Highway.

(c) Effect of Radio Alert
The radio alert changes the disruption probability of R1 from 0.30 to 0.80.
Therefore:
P(Disruption) = 0.80

P(Normal) = 1 − 0.80 = 0.20

Recalculate R1 Expected Utility

EU(R1) = (0.20 × 70) + (0.80 × 10)
EU(R1) = 14 + 8
EU(R1) = 22
Therefore:
EU(R1) = 22
The expected utilities of the other routes remain:
EU(R2) = 51.5
EU(R3) = 50

New Expected Utility Table

Route	New Expected Utility
R1 – Highway	22
R2 – City	51.5
R3 – Ring Road	50

Therefore, after receiving the radio alert:
R2 – City has the highest expected utility.
New Rational Choice
The rational choice changes from R1 to R2.

(d) Actual Outcome and Rationality

Suppose the agent selected R1 – Highway in part (b), because it had the highest expected utility.
If a disruption occurs, the travel time becomes:
T = 70 minutes
Since 70 > 55:
L = 1
Therefore:
U = 100 − 70 − 20(1)
U = 10
So, the actual utility is:
Actual Utility = 10

Was the agent irrational?

No, the agent was not irrational.
At the time of making the decision, the agent used the available probability information:
P(disruption) = 0.30
R1 had the highest expected utility:
EU(R1) = 52
Therefore, choosing R1 was rational based on the information available at that time.
A rational agent does not need to guarantee the best actual outcome. It chooses the action with the highest expected performance based on its available information.
An omniscient agent, however, would know in advance that the disruption would occur and could choose another route.

(e) Interpretation

Rationality depends on the percept sequence, available information, possible actions, and performance

measure.

Initially, the agent believed that R1 had a disruption probability of only 0.30. Therefore, R1 had the highest expected utility:

EU(R1) = 52

After receiving the radio alert, the disruption probability increased to 0.80.

This changed the expected utility of R1 to:

EU(R1) = 22

Therefore, the agent changed its decision to R2, whose expected utility remained:

EU(R2) = 51.5

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____